

discusses how a Wi-Fi router can be used to provide far-field wireless power for gadgets. In their first prototype, for the first time in the world, they have demonstrated how Wi-Fi chipsets can power camera sensors or li-ion coin-cell batteries from more than 6m (20-feet) away. Wi-Fi receivers had so far been used to capture information from Wi-Fi radio broadcasts. The research team has sought a way to harvest energy from these broadcasts.

The engineers connected an antenna to a temperature sensor and put it near a Wi-Fi router so that voltages in the device could be measured to determine the time for which the device could operate on a remote power source. To make things work, they programmed these devices for broadcasting continuous power to an energy-harvesting sensor. The results showed that a temperature sensor could operate at a distance of up to six metres.

Pocket-size drone that can fold up

Researchers have designed a small, foldable drone, inspired by origami, which can unfold itself automatically and take flight within a fraction of a second. A large number of these quadcopters, the size of an outstretched palm, could be released over a disaster zone to take photographs and make contact with survivors.

When the device is not in use, the arms, which are made of fibre glass and light, yet rigid polyester, fold up into a trapezoid. When switched on, force of the propellers causes the arms to unfold horizontally. Then, magnets keep the arms locked into position.

For the drone to remain stable during flight, two propellers (diagonally across from each other) spin in opposite directions from the other two. Propellers all spin in the same direction at first while the drone unfolds, but a sensor detects when the arms have locked into position, and within 50 milliseconds, direction of two of the rotors is reversed, and the drone is ready to take flight.

World's first electronic multi-state memory cell

Researchers at RMIT University's MicroNano Research Facility (MNRF), Australia, have built the world's first electronic multi-state memory cell that can mirror a brain's ability to simultaneously process and store multiple strands of information.

This brings them closer to imitating key electronic aspects of the human brain, which is an important step towards creating a bionic brain. This could help unlock successful treatments for common neurological conditions such as Alzheimer's and Parkinson's.

Project leader Dr Sharath Sriram, co-leader of RMIT Functional Materials and Microsystems Research Group, has said that the ground-breaking development imitates the way the brain uses long-term memory.

The research builds on RMIT's previous discovery

where ultra-fast nano-scale memories were developed using a functional oxide material in the form of an ultra-thin film, which is 10,000 times thinner than a human hair.

Google, Levi Strauss to launch smartclothes

Google is all set to tie up with popular jean maker Levi Strauss to launch smartclothes using particular woven fabric with touchscreen-control capabilities. Named Project Jacquard, the plan would be implemented by a small team at Google called Advanced Technology and Projects (ATAP).

The project is named Jacquard after a Frenchman who invented a type of loom. While the clothes are expected to be stretchable and washable like normal fabric, these would also be able to connect with devices. Special threads would be woven into a wide array of fabrics. However, conductivity will be limited to desired parts of the fabric or spread across entire cloth.

Google has said that, with the use of standard, industrial looms, touch and gesture interactivity could be woven to any textile. Hence, anything involving fabric is likely to have computer touchpad-style control capabilities woven into it.

ATAP also said that the conductive yarn would be connected to minute circuits, no bigger than jacket buttons and small electronics that can use algorithms to recognise touch or swipes.

Daimler, Qualcomm to develop in-car tech

Car maker Daimler and Qualcomm Inc. have partnered to develop wireless recharging of mobile phones in cars as well as recharging of electric cars without cables.

The two companies are assessing the application of wireless technology to charge their electric vehicles (EVs) and plug-in hybrid EVs without having to plug these in. They are also exploring technologies that will enable customers to wirelessly charge devices such as mobile phones while driving their cars, as well as ways to enhance in-car experience through high-speed 3G/4G connectivity.

Dyson CSYS LEDs to stay bright for 37 years

LEDs are meant to last for a long time, but if not properly heat shielded, these tend to lose their brightness and colour. Jake Dyson of Jake Dyson Light has designed CSYS LEDs that will stay bright for 37 years.

Conventional applications fail to protect LEDs from heat and are often subjected to temperatures up to 130°C. The heat has a damaging effect on the phosphorous coating of the bulb, which results in degradation in brightness and colour over time.

CSYS task lights have been engineered to use heat pipe technology that takes away the heat from the LEDs and keeps them cool at around 55°C. The lower temperature allows the lights to stay bright for 37 years.